

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-28 09:45 UTC

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Generated 2026-04-28 09:45 UTC

- Cycles recorded: **183**
- Time span: 108.1h (~1.69 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	160
FALSIFIED	14
BARRIER_HIT	5
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier analysis of boolean functions	3
Noncommutative geometry	2
Representation theory of symmetric groups	2
Ergodic Circuit Framework (COMM_COMPLEXITY)	2
Matroid Theory	2
Algebraic Topology	2
Tropical geometry	2
Random Matrix Theory	2
Combinatorial homotopy theory	1
Kazhdan-Lusztig theory of Hecke algebras	1
Combinatorial algebraic topology	1
Knot theory (quantum invariants)	1
Quadratic forms over finite fields (Grothendieck-Witt groups)	1
Clifford algebras over finite fields	1
Toric geometry (via Gröbner degenerations)	1

Field	Cycles
Arithmetic geometry (Tate-Shafarevich groups)	1
Geometric Langlands correspondence	1
Algebraic graph theory (chromatic polynomials)	1
Clifford algebras over real vector spaces	1
Algebraic cycles (Chow groups over finite fields)	1
Algebraic lattice theory (Möbius functions of flow lattices)	1
Directed algebraic topology (d-space homology)	1
Motivic integration (Denef-Loeser zeta functions)	1
Matroid theory (Clifford index of binary matroids)	1
Representation theory of finite groups (Frobenius-Schur indicator)	1
Categorification and knot Floer homology	1
Modular forms	1
Algebraic geometry (ideals in polynomial rings)	1

Last 15 cycles (chronological)

Time	Verdict	Title
2026-04-27 20:42 UTC	BARRIER_HIT	Median-Restriction Hardness Amplification Gap for DNF Truth Table
2026-04-27 21:18 UTC	INCONCLUSIVE	Total-Influence Defect of Clause-Indicator Bounds Tree-Resolution
2026-04-27 21:34 UTC	BARRIER_HIT	Diameter-Multiplicity Power Law under Controlled Refinement Recov
2026-04-27 21:48 UTC	INCONCLUSIVE	Commutator-Length Defect in S_5 Predicts Width-5 BP Size of Boole
2026-04-27 22:23 UTC	INCONCLUSIVE	Hessian-Rank of Clause-Count Polynomial Bounds DPLL Depth
2026-04-27 22:52 UTC	INCONCLUSIVE	Herbrand-Disjunction Length of PHP Bounds
2026-04-27 23:24 UTC	INCONCLUSIVE	Tree-Resolution Leaves Beck-Fiala Slack of Clause-Variable Hypergraph Bounds DPLL Leaves
2026-04-27 23:36 UTC	BARRIER_HIT	Monotone Ultralimit Convergence of κ along the Asymptotic Complet
2026-04-27 23:55 UTC	INCONCLUSIVE	Spectral-Tail Mass of Unit-Propagation Indicator Bounds DPLL Leav
2026-04-28 01:39 UTC	INCONCLUSIVE	Block-Composition Sub-Additivity of Coarse Depth κ via the Roe-Tr

Time	Verdict	Title
2026-04-28 03:39 UTC	INCONCLUSIVE	Coboundary-Vanishing Threshold: HX^1 Triviality Below the Diamete
2026-04-28 08:37 UTC	INCONCLUSIVE	Multiplicity-Doubling Lower Bound from Protocol Pullback on Index
2026-04-28 08:46 UTC	INCONCLUSIVE	Σ^b_0 -PIND Width-2 Closure Round-Count vs DPLL Leaves
2026-04-28 09:18 UTC	BARRIER_HIT	Mahler Measure of Truth-Table Polynomial Lower-Bounds DNF_min
2026-04-28 09:45 UTC	INCONCLUSIVE	Plünnecke-Ruzsa Doubling of Constraint Vectors Lower-Bounds SoS D

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```